XAU/USD & ETH/USD AI Prediction System — Project Document

Advanced Financial Market Analysis Platform

Project Snapshot

An enterprise-grade algorithmic trading and market-analysis platform that combines deep learning and traditional technical analysis to forecast short-term (20-minute) price movements for XAU/USD (gold) and ETH/USD (Ethereum). Built for real-time inference, rigorous backtesting, and production-grade signal generation with integrated risk management.

Core Capabilities

- **Short-term forecasting:** 20-minute horizon predictions with real-time inference.
- **Hybrid model ensemble:** LSTM deep learning core with XGBoost for validation and confidence scoring.
- **Multi-source data pipeline:** Market data from TwelveData, Binance, and local historical datasets with robust fallback and caching.
- **Automated signal generation:** Trade signals with stop-loss/take-profit, position sizing, and threshold filtering.
- **Interactive dashboard:** Streamlit UI with candlesticks, forecast overlays, and performance tracking.

System Architecture

Frontend: Streamlit dashboard for monitoring, visualization, and manual controls.

Modeling Layer:

- Primary: LSTM sequence model (2 stacked LSTM layers, dropout) trained on 60-step sequences.
- Secondary: XGBoost regressor using technical indicators and engineered features to provide ensemble validation and confidence scores.

Data & Inference Pipeline:

- Multi-source ingestion with three-tier fallback (TwelveData → Binance → local CSV).
- Smart caching (in-memory with configurable TTL) and automatic refresh.
- Preprocessing: scaling (MinMax), feature engineering (technical indicators, temporal features), and sequence generation for LSTM.

Serving:

- Real-time inference optimized with TensorFlow @tf.function and low-latency execution for sub-3s per-asset predictions.
- Model files managed via Git LFS with compressed artifacts for distribution.

Feature Engineering & Indicators

- Price series (OHLC + volume), momentum indicators (RSI, MACD, Stochastic), trend indicators (EMA, ADX), volatility measures (Bollinger Bands, ATR), and volume metrics (VWAP, OBV).
- Temporal features to capture intraday seasonality and periodic patterns.

Prediction Engine & Signal Logic

- **Forecasting window:** 20 minutes ahead per asset.
- **Ensemble confidence:** Blend LSTM output with XGBoost score; apply threshold filters and multi-confirmation rules before emitting BUY/SELL/WAIT signals.
- **Risk management:** Dynamic position sizing, stop-loss/take-profit calculations, maximum daily trade limits, and minimum movement thresholds to mitigate false positives.

Performance & Backtesting

- **Target latency:** <3 seconds inference per asset; refresh cadence configurable (typical 5 minutes).
- **Backtesting:** Rolling-window validation and historical simulation; reported accuracy metrics per asset and timeframe.
- **Data requirements:** Minimum 2,000 historical points recommended for stable modeling.

Deployment & Operations

- **Runtime requirements:** Python 3.9+, TensorFlow 2.13+, XGBoost, Redis-like caching, and Streamlit for dashboards.
- **Hardware:** 4+ GB RAM (8 GB recommended), 4 CPU cores; GPU optional for faster training.
- **Model lifecycle:** Versioned artifacts (bz2 compression), scaler serialization (pickle), and reproducible training scripts.
- **Scheduling:** Configurable schedule runner (PKT / UTC+5): daily runs at 05:00 and 17:00 PKT plus continuous 20-minute monitoring.

Security & Access Control

- Credential-based access with optional IP whitelisting and subscription-based gating.
- API keys and secrets kept in environment variables; no long-term sensitive storage in plaintext.
- Rate limiting for external APIs and robust exception handling across ingestion and inference.

Development Workflow & Testing

- Unit tests for preprocessing, model inference, and utility functions.
- Integration tests for data fetching and backtesting pipelines.
- Performance tests for end-to-end latency and throughput under simulated loads.

Visualization & UX

- Real-time candlestick charts with prediction overlays, interactive zoom/pan, and timeframe selection.
- Signal dashboard with color-coded indicators, risk metrics, and historical performance charts.
- Exportable visual reports (PNG/PDF) and downloadable model/metrics artifacts for auditability.

Risk Controls & Safeguards

- Minimum movement thresholds, ATR-based volatility filters, liquidity checks, and news-awareness hooks to avoid trading during high-impact events.
- Position sizing and daily trade caps to limit exposure.
- Logging, alerting, and graceful failure handling for operational resilience.

Business Value

- Accelerates decision-making by delivering high-frequency, model-driven signals for precious metals and crypto markets.
- Reduces manual monitoring costs and standardizes risk-aware execution logic.
- Suitable for quant research teams, prop desks, and systematic trading operations.

Quick Start (Developer)

1. Set up Python 3.9+ virtual environment and install dependencies.
2. Configure API keys: `TWELVEDATA_API_KEY`, optional `BINANCE_API_KEY`/ `BINANCE_SECRET_KEY`.
3. Place models and artifacts under `/models/` and historical data under `/data/`.
4. Launch dashboard: `streamlit run app.py --server.port=8501`.

Recommended packages: tensorflow, xgboost, pandas, numpy, streamlit, plotly, ta-lib or pandas-ta, chardet.

**Prepared for professional presentation — no personal contact details included. **

It's accuracy is not perfect as it display because there is need more training models for good accuracy so due to low capability and capacity this model was trained at that maximum by me